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Ekman Spirals with Variable Eddy Viscosity

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Contents

1 Background

1.1 Introduction

The Ekman spiral is a famous consequence of the Coriolis force in meteorology and oceanography. Due to the interaction between shear forces in the atmosphere and ocean, the wind high up in the atmosphere, and current in the deep ocean, will not give rise to a flow in the same direction at the surface. The adjustment of the flow from the surface will thus spiral, and is therefore properly named the Ekman spiral after its discoverer Ekman [1905]. The surface flow is often at an angle from the flow aloft (or down under in the ocean) by an angle of more or less 45° [Ekman, 1905]. Due to this turning of the wind, the net flux of the wind will be perpendicular to the wind aloft (or current below). This is known as the Ekman transport and is crucial for studying both meteorology and oceanography.

However, the wind turning 45° assumes a constant viscosity in the medium. This is a unrealistic when a turbulent boundary layer is present. **DISCUSS TURBULENT WIND.** Thus, in reality the viscosity should vary with altitude. This would change the wind-turning from 45°, and thus also the Ekman transport. In this article, a varying viscosity with altitude will be studied under the Ekman equation. This will be done both analytically and numerically and will also be compared to model data. This will give a deeper insight into the Ekman spiral and provide insight into when a surface angle different from 45° can be observed.

2 Theory

2.1 Problem formulation

The Ekman equation is derived from the steady-state Navier–Stokes equations on a rotating frame with shear flow. Assuming friction, incompressible flow, negligible vertical motion, gives the horizontal momentum equations as

$$\begin{cases} -fv = -\dfrac{1}{\rho}\dfrac{\mathrm{d}p}{\mathrm{d}x} + \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}u}{\mathrm{d}z}\right), & \text{(1a)} \\ fu = -\dfrac{1}{\rho}\dfrac{\mathrm{d}p}{\mathrm{d}y} + \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}v}{\mathrm{d}z}\right), & \text{(1b)} \end{cases}$$

where u and v are the horizontal velocity components, f is the Coriolis parameter, ρ is the fluid density, and ν is the viscosity. Far away from the surface, shear flow and friction becomes negligible and the flow will thus approach geostrophic balance, $(u, v) \to (u_g, v_g)$, where one can neglect the friction term in Equation 1b. Subtracting the geostrophic balance from Equation 1b gives

$$\begin{cases} -f(v - v_g) = \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}u}{\mathrm{d}z}\right), & \text{(2a)} \\ f(u - u_g) = \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}v}{\mathrm{d}z}\right). & \text{(2b)} \end{cases}$$

This problem can be formulated with one equation if one introduces a complex velocity as

$$U \equiv u + iv, \tag{3}$$

and a complex geostrophic velocity as $U_g = u_g + iv_g$. Using Equation 2b with the complex velocities U and U_g the coupled equations can be combined into the Ekman equation

$$\frac{\mathrm{d}}{\mathrm{d}z}\left(\nu\frac{\mathrm{d}U}{\mathrm{d}z}\right) = if(U - U_g). \tag{4}$$

With the boundary conditions

$$U(0) = 0, \tag{5a}$$

$$U(z \to \infty) = U_g. \tag{5b}$$

One can determine the surface wind-turning angle by taking the argument of the infinitesimal change of the wind at the ground,

$$\theta = \arg\left(\left.\frac{\mathrm{d}U}{\mathrm{d}z}\right|_{z=0}\right). \tag{6}$$

The Ekman transport is defined as the mass flux per unit width for the entire Ekman spiral. One can thus define the Ekman transport as

$$T = \int_0^\infty [U - U_g]\,\mathrm{d}z, \tag{7}$$

since the transport represents the net movement of the entire fluid column, offset from the geostrophic wind aloft. One can consequently determine the angle of the Ekman transport by taking the argument, $\theta_T = \arg(T)$.

2.2 The vertical momentum flux

The Ekman equation can be reformulated by introducing a complex vertical momentum flux as

$$\tau = \nu\frac{\mathrm{d}U}{\mathrm{d}z}. \tag{8}$$

Using this definition, Equation 4 can be rewritten as

$$\boxed{\frac{\mathrm{d}^2\tau}{\mathrm{d}z^2} = i\underbrace{\lambda}_{\equiv\frac{f}{\nu}}\tau.} \tag{9}$$

Here, an eigenvalue λ has been introduced to the system. The new eigenvalue form is more intuitive since it directly compares the second order derivative with the variable itself. Furthermore, Equation 9 is comparable with famous equations such as the Helmholtz equation and the time-independent Schrödinger equation, where λ becomes analogous to the potential. The new equation also has a clear physical interpretation, as it describes the vertical structure of the momentum flux in the Ekman problem. Using Equation 9, the boundary conditions become

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=0} = -ifU_g, \tag{10a}$$

$$\tau(z\to\infty) = 0. \tag{10b}$$

This new quantity will furthermore not change the surface angle, as the viscosity is always a positive real parameter and thus does not affect the argument,

$$\theta = \arg\left[\tau(z=0)\right]. \tag{11}$$

Solving for the Ekman transport becomes easier by using the momentum flux, as it directly reduces the integral of the Ekman transport. By observing the Ekman equation (Equation 4) one can observe $\frac{\mathrm{d}\tau}{\mathrm{d}z} = -if\,(U - U_g)$. By using the boundary condition that the momentum flux vanishes as infinity, the Ekman transport reduces to

$$T = \int_0^\infty [U - U_g]\,\mathrm{d}z = \frac{i}{f}\tau(z=0), \tag{12}$$

which is proportional to the vertical momentum flux at the surface. Consequently, this can be interpreted as the surface stress at the ground being proportional to the Ekman transport of the fluid column.

2.3 The classical solution

The classical solution to the Ekman equation is obtained when the viscosity is assumed to be constant with altitude, $\nu =$ constant, and can be found in Ekman [1905]. In this case the classical Ekman spiral solution is

$$\tau(z) = \nu U_g\frac{1+i}{h_{\mathrm{Ek}}}e^{-\frac{1+i}{h_{\mathrm{Ek}}}z}, \tag{13}$$

where

$$h_{\mathrm{Ek}} \equiv \sqrt{\frac{2\nu}{f}} \tag{14}$$

is the Ekman depth scale. The surface angle of the classical Ekman theory with constant viscosity gives 45°, since the solution is equally real and imaginary at $z = 0$. The transport yields an angle of 90° from the surface wind-turing, meaning that the transport and surface wind is perpendicular.

2.4 The two layer model

The next natural step from a constant viscosity profile is to assume two layers of constant viscosity in each layer. One can label the lower level as having the viscosity ν_1 up until a height of H, where the next viscosity of ν_2 takes over. From this assumption one would obtain a superimposed solution similar to Equation 13 as

$$\begin{cases} \tau_1(z) = A_1 e^{\frac{1+i}{h_{\mathrm{Ek1}}}z} + B_1 e^{-\frac{1+i}{h_{\mathrm{Ek1}}}z}, & (15a) \\ \tau_2(z) = B_2 e^{-\frac{1+i}{h_{\mathrm{Ek2}}}z} & (15b) \end{cases}$$

where the Ekman depth scales are defined for each layer like in Equation 14. To solve for the constants above one has to use the fact that the velocity and momentum flux has to be continuous, which gives the jump condition

$$U_1 = U_2 \quad \text{at } z = H, \tag{16a}$$

$$\tau_1 = \tau_2 \quad \text{at } z = H. \tag{16b}$$

Using this and the boundary conditions at Equation 5 one obtains a expression between for factors as

$$A_1 = -\nu_1 U_g \frac{1+i}{h_{\mathrm{Ek1}}} \cdot \frac{1-\gamma}{1-\gamma+e^{2(1+i)\varphi}(1+\gamma)}, \text{ where } \gamma = \sqrt{\tfrac{\nu_2}{\nu_1}} \tag{17a}$$

$$B_1 = \nu_1 U_g \frac{1+i}{h_{\mathrm{Ek1}}} \cdot \frac{(1+\gamma)\, e^{2(1+i)\varphi}}{1-\gamma+e^{2(1+i)\varphi}(1+\gamma)}, \tag{17b}$$

$$B_2 = \nu_2 U_g \frac{1+i}{h_{\mathrm{Ek2}}} \cdot \frac{2e^{(1+i)\varphi\left(1+\gamma^{-1}\right)}}{1-\gamma+e^{2(1+i)\varphi}(1+\gamma)}, \tag{17c}$$

where φ is the non-dimensional layer thickness defined as

$$\varphi = \frac{H}{h_{\mathrm{Ek1}}}. \tag{18}$$

This gives a full expression for the vertical momentum flux in the two layer model and can be seen in Figure 1. One can observe how vertical momentum flux changes with altitude for different viscosity ratios and non-dimensional heights, φ. The

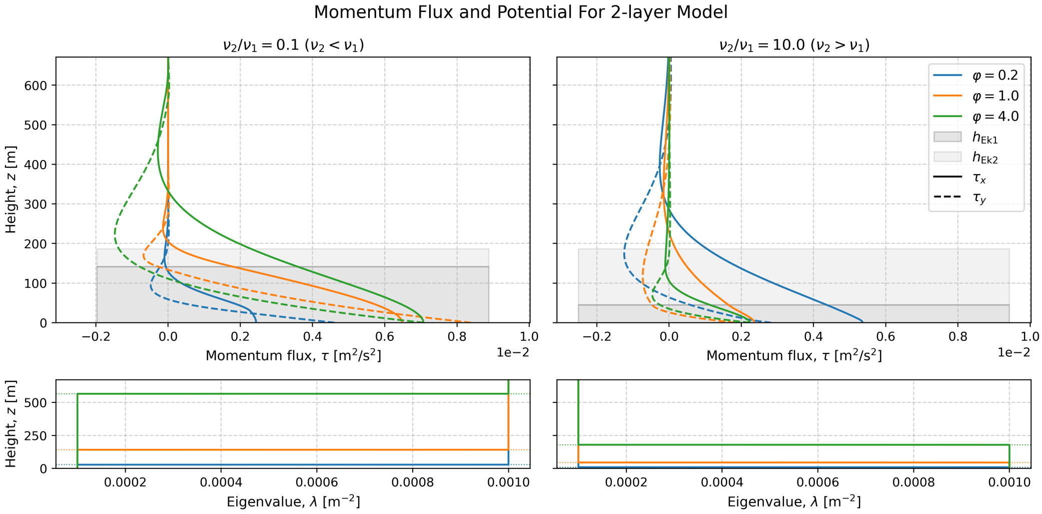

Figure 1: The vertical structure of the momentum flux for the two layer model. The momentum flux is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek1}}$ and the ratio of the viscosities. The corresponding Ekman heights and eigenvalue is shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow and a Coriolis parameter of $f = 1 \times 10^{-4}\,\mathrm{s}^{-1}$.

Using the constant R

$$R = \frac{1-\gamma}{1+\gamma}. \tag{19}$$

and the derived solution for the vertical momentum flux, one can obtain the surface wind-turning angle as

$$\theta = \arg\left[\tau(z=0)\right] = \arg\left[(A_1 + B_1)\right] \tag{20}$$

$$= 45^\circ + \arg\left[\frac{e^{(1+i)\varphi} - Re^{-(1+i)\varphi}}{e^{(1+i)\varphi} + Re^{-(1+i)\varphi}}\right], \tag{21}$$

which an be observed in Equation 21. There one can observe solutions for different viscosity ratios, half of which have an higher upper-level viscosity and the other half a lower one. These are plotted together with a 45° reference point to highlight the difference from the classical Ekman solution. One can observe a convergence towards the classical Ekman value when $\varphi \to \infty$ for all the viscosity ratios. The zero limit is, however, dependent on the viscosity ratio, where finite ratios approach 45°. However, when $\frac{\nu_2}{\nu_1} \to 0$ and $\varphi \to 0$ one can observe that $\theta \to 90°$. Respectively, one can observe that when $\frac{\nu_2}{\nu_1} \to \infty$ and $\varphi \to 0$, $\theta \to 0°$. For $\nu_1 > \nu_2$ (and $\nu_2 > \nu_1$ respectively) the solution will always provide a result greater than 45°, except between the $\varphi \in \left(\frac{\pi}{2}, \pi\right)$ were a angle below 45° can be observed.

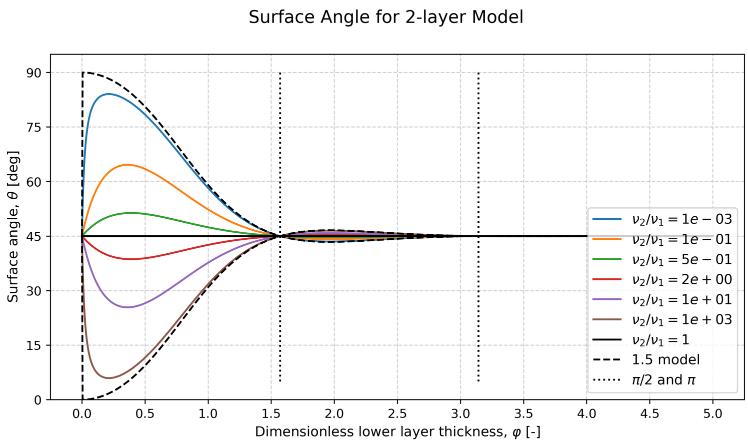

Figure 2: The surface angle as a function of the dimensionless layer thickness $\varphi = H/h_{\mathrm{Ek1}}$ for the two-layer viscosity model. Different coloured curves correspond to different layer viscosity ratios. The classical Ekman angle of 45° is shown as a reference.

2.5 The exponential decaying viscosity

Another viscosity profile that can provide a fully analytical solution is an exponentially decaying viscosity with altitude. Here the viscosity is set to decay from an initial viscosity of ν_0, with a decay rate of k as $\nu(z) = \nu_0 \exp(-kz)$. To solve Equation 9 with this viscosity profile one can introduce the new variable

$$t = 4\,(1+i)\ \varphi_{\text{exp}}\, e^{\frac{kz}{4}}. \tag{22}$$

The non-dimensional parameter φ_{exp} resembles the non-dimensional layer thickness from the two-layer model, but is here defined as

$$\varphi_{\text{exp}} = \frac{1}{k\,h_{\text{Ek0}}} = \frac{1}{k}\sqrt{\frac{f}{2\nu_0}}. \tag{23}$$

Using these newly defined variables Equation 9 transforms into

$$t^2\frac{\mathrm{d}^2\tau}{\mathrm{d}t^2} + t\frac{\mathrm{d}\tau}{\mathrm{d}t} - t^2\tau = 0, \tag{24}$$

which is a differential equation with a standard solution. The standard solution is a modified Bessel equation as

$$\tau(t) = C_1 I_0(t) + C_2 K_0(t), \tag{25}$$

where C_1 and C_2 are constants, while $I_0(t)$ and $K_0(t)$ are modified Bessel functions of the first and second kind of order zero, respectively. By applying the previously defined boundary conditions from Equation 10 one can find the two constants. Applying the geostrophic boundary condition from Equation 10b one can see that $C_1 = 0$ since $I_0(t)$ does not go to zero when $t \to \infty$ (which occurs when $z \to \infty$). Applying the surface boundary condition from Equation 10a one can find an expression for the second constant. By transforming the solution to depend on the variable z one can obtain the solution

$$\tau(z) = \frac{(1+i)\,fU_g}{2k\varphi}\frac{K_0\left(4\,[1+i]\,\varphi_{\text{exp}}e^{kz/4}\right)}{K_1\left(4\,[1+i]\,\varphi_{\text{exp}}\right)} \tag{26}$$

This full solution for the vertical momentum flux in the exponential model can be observed in Figure 1. There one can observe how vertical momentum flux changes with altitude for different non-dimensional heights, φ.

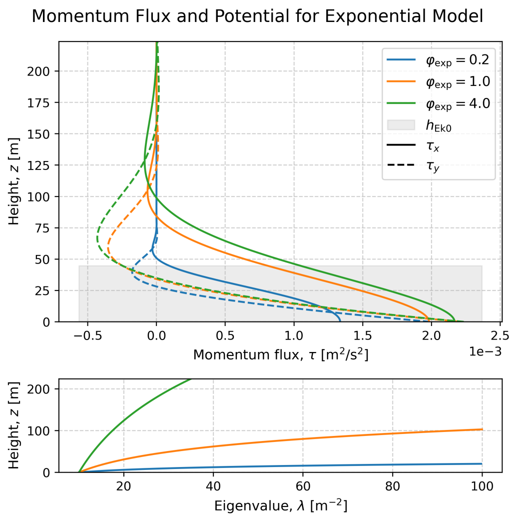

Figure 3: The vertical structure of the momentum flux for exponentially decreasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi_{\text{exp}} = 1/kh_{\text{Ek0}}$. The corresponding Ekman heights and eigenvalue is shown with the shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow and a Coriolis parameter of $f = 1 \times 10^{-4}\,\text{s}^{-1}$

Using Equation 26 one can obtain the surface angle of this model as

$$\theta = \arg\left[\tau(z=0)\right] = 45^\circ + \arg\left[\frac{K_0\left(4\,[1+i]\ \varphi_{\text{exp}}\,\right)}{K_1\left(4\,[1+i]\ \varphi_{\text{exp}}\,\right)}\right], \tag{27}$$

where K_1 is the modified Bessel function of the second kind of order one.

Figure 4 showcases the result from the exponentially decaying viscosity as in Equation 27. There one can observe a similar limit as in Figure 2 when $\varphi \to 0$ for $\nu_2 = 0$, as Equation 27 also approaches 90°. The same limit is also true for $\varphi \to \infty$, which gives $\theta \to 45°$, although this exponential solution converges slightly slower than the two-layer solution. However, the shape is much more uniform, with an exponential-like decay. All angels are greater then the reference angle of 45°.

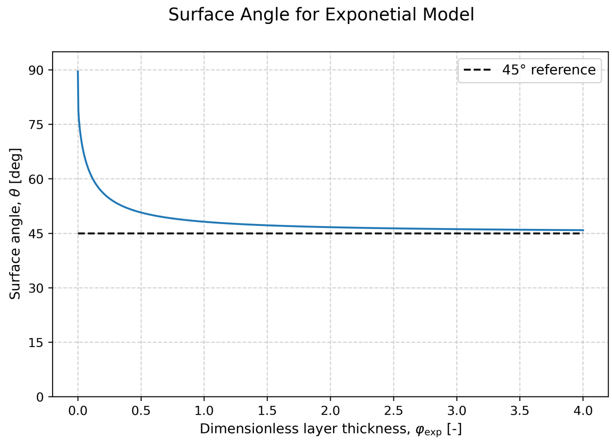

Figure 4: Surface angle for the exponentially decaying viscosity model as a function of the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}}$. The solution assumed a purely zonal geostrophic flow.

2.6 The simplified upper boundary condition model

In the previous sections two completely analytical solutions for two different altitude-varying viscosity profiles were derived. The difficulty in solving Equation 9 comes from the altitude dependence of the viscosity profile $\nu(z)$. This introduces variable coefficients into the complex second-order ordinary differential equation, which in general prevents closed-form analytical solutions. Analytical solutions therefore exist only for specific viscosity profiles that allow the equation to be transformed into a solvable form.

Equation 9 is furthermore also difficult to solve in a numerical setting, as the upper boundary condition is a limit. This will put limits on the viscosity profiles one can study, as the viscosity cannot increase with altitude continuously. In the physical world, this makes sense, since we assume that high up in the atmosphere (or deep down in the ocean), the flow becomes laminar and the turbulent viscosity small. This was what was labelled as the geostrophic limit. By limiting ourselves to only decreasing viscosity profiles, the differential equation becomes difficult to solve. Furthermore, if one wants to solve for a dimensionless height parameter, too complex viscosity profiles become hard to deal with.

A solution to this problem is to introduce the simplified upper boundary condition model. For the two-layer model, this is reduced to a 1.5 layer model, where we assume that the upper-level viscosity is negligible $\nu_2 \approx 0$, which could be observed as the limit in Figure 2. This also makes physical sense, since we know that both the atmosphere and the ocean consist of a boundary layer where turbulent mixing takes place. Allowing the viscosity to vary freely in the lower layer reduces the problem to a simplified upper boundary condition model. This will allow viscosity profiles that do not converge at $z \to \infty$, since the new domain is between $z \in [0, H]$. This modifies the boundary conditions in Equation 10 as

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=0} = -ifU_g, \tag{28a}$$

$$\tau(z = H) = 0. \tag{28b}$$

Above, H becomes the height of the boundary layer, and the upper boundary condition was argued from the jump condition in the two-layer model, as seen in Equation 16b.

2.6.1 Simple viscosity profiles for the SUBC model

Using the the simplified upper boundary condition model, SUBC model, one can study some simple viscosity profiles to be evaluated numerically. The easiest, except for the constant viscosity, is a linearly varying viscosity with height as

$$\nu_{\uparrow}(z) = \frac{\nu_{\max}}{H} z, \tag{29a}$$

$$\nu_{\downarrow}(z) = \frac{\nu_{\max}}{H} (H - z), \tag{29b}$$

where $\nu_{\max}$ is the maximal viscosity. For both of these profiles one can define a dimensionless height coordinate as

$$\tilde{z} = \frac{z}{H} \implies \begin{cases} \nu_{\uparrow}(z) = \nu_{\max}\tilde{z} \\ \nu_{\downarrow}(z) = \nu_{\max}(1 - \tilde{z}). \end{cases} \tag{30}$$

One must note that this new coordinate changes the previous z-derivative, as a factor of $1/H$ appears when taking the derivative with respect to $\tilde{z}$. Using these linear viscosity profiles, together with the new dimensionless height coordinate, one can rewrite Equation 9 for the two profiles as

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{\tilde{z}}}_{\equiv 1/\tilde{\nu}_{\uparrow}} \underbrace{\frac{Hf^2}{\nu_{\max}}}_{=2\varphi^2} \tau, \tag{31a}$$

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{1-\tilde{z}}}_{\equiv 1/\tilde{\nu}_{\downarrow}} \underbrace{\frac{Hf^2}{\nu_{\max}}}_{=2\varphi^2} \tau. \tag{31b}$$

Above the dimensionless height φ was again introduced following the logic from before.

Another simple profile to evaluate is a parabolic increasing viscosity as

$$\nu_p = \frac{4\nu_{\max}}{H^2} (H - z) z. \tag{32}$$

Again, using the non-dimensional height coordinate $\tilde{z}$ and the dimensionless height φ, one can rewrite Equation 9 as

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{4\tilde{z}\left[1-\tilde{z}\right]}}_{\equiv 1/\tilde{\nu}_p} \underbrace{\frac{Hf^2}{\nu_{\max}}}_{=2\varphi^2} (U - U_g). \tag{33}$$

3 Methodology

3.1 Numerical method

For arbitrary viscosity profiles, Equation 4 generally does not admit a closed-form analytical solution. The equation was therefore solved numerically as a boundary value problem using the `solve_bvp` routine available in `SciPy` [Virtanen et al., 2020].

For the SUBC model, it was shown that the equation could be rewritten for a dimensionless layer thickness φ. Thus, using an arbitrary height-normalized height-dependent eddy viscosity profile, $\tilde{\nu}(z)$, one can write a simpler differential equation. Separating the equation into the zonal and meridional velocity components, one obtains

$$\frac{\mathrm{d}^2\tau_{\mathrm{x}}}{\mathrm{d}z^2} = -\frac{1}{\tilde{\nu}(z)} 2\varphi^2 \tau_y, \tag{34a}$$

$$\frac{\mathrm{d}^2\tau_{\mathrm{y}}}{\mathrm{d}z^2} = \frac{1}{\tilde{\nu}(z)} 2\varphi^2 \tau_x \tag{34b}$$

Here it was also assumed that the wind aloft was set as a purely zonal u_g. To apply the numerical boundary value solver, the equations were rewritten as a first-order system by introducing the new variables

$$\tau'_x = \frac{\mathrm{d}\tau_x}{\mathrm{d}z}, \qquad \tau'_y = \frac{\mathrm{d}\tau_y}{\mathrm{d}z}. \tag{35}$$

Thus one can expand Equation 34b into four coupled first-order equations

$$\begin{cases} \frac{\mathrm{d}\tau_x}{\mathrm{d}z} &= \tau'_x, \\ \frac{\mathrm{d}\tau'_x}{\mathrm{d}z} &= -\frac{2\varphi^2\tau_y}{\tilde{\nu}(z)}, \\ \frac{\mathrm{d}\tau_y}{\mathrm{d}z} &= \tau'_y, \\ \frac{\mathrm{d}\tau'_y}{\mathrm{d}z} &= \frac{2\varphi^2\tau_x}{\tilde{\nu}(z)} \end{cases} \tag{36}$$

with the boundary conditions

$$\tau'_x(\varphi = 0) = 0, \qquad \tau_x(\varphi = 1) = 0, \tag{37a}$$

$$\tau'_y(\varphi = 0) = -fu_g, \qquad \tau_y(\varphi = 1) = 0. \tag{37b}$$

If the viscosity profile, $\tilde{\nu}$, obtains a value of zero in the domain, then the coupled Equation 36 becomes unsolvable since we divide by $\hat{\nu} = 0$. In this scenario one has to introduce a constant background viscosity, ϵ as

$$\hat{\nu} \to \hat{\nu} + \epsilon, \tag{38}$$

where $\hat{\nu}/\epsilon \gg 1$ was assumed. This background viscosity was thus applied for the linear increasing model and the parabolic model since both have zero viscosity at the surface.

The SUBC model equations were solved iteratively on a vertical grid. The solver required an initial guess, which was chosen based on the initial setup of the system. The initial guess was therefore set as the classical solution for a constant viscosity. The final solution was found when the iterative solution reached a steady state for the differential equations with the imposed boundary conditions.

4 Analysis

4.1 The SUBC model

4.1.1 The vertical structure of the momentum flux

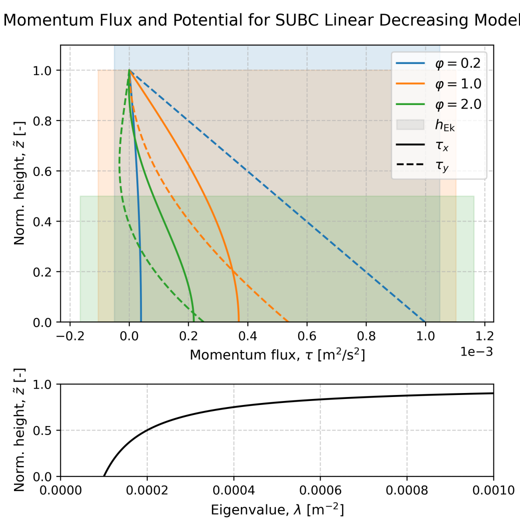

Figure 5: Numerically computed vertical structure for linearly decreasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}} = 1/\tilde{h}_{\mathrm{Ek}}$. The corresponding Ekman heights and eigenvalue is shown with the shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow.

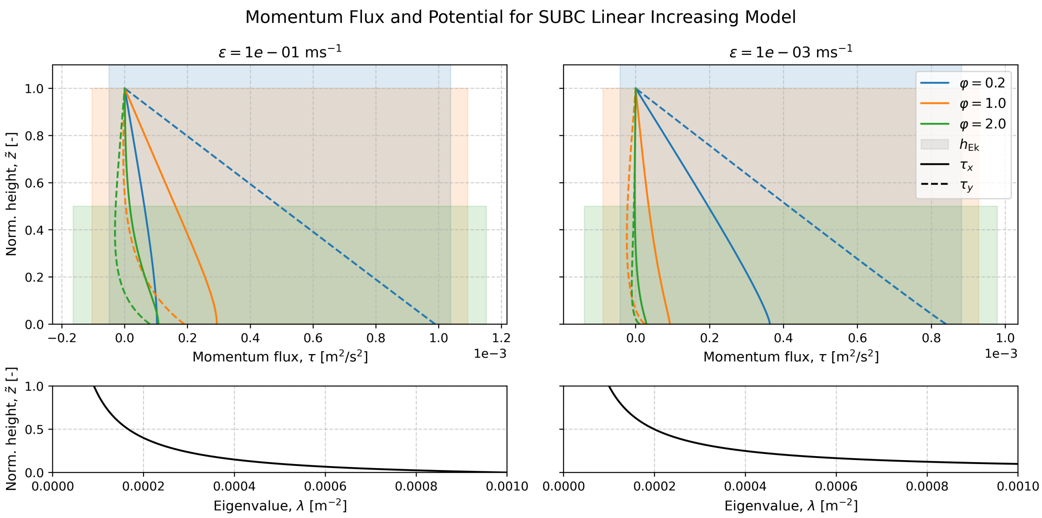

Figure 6: Numerically computed vertical structure for linearly increasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}} = 1/\tilde{h}_{\mathrm{Ek}}$ and the background viscosity ϵ. The corresponding Ekman heights and eigenvalue is shown with the shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow.

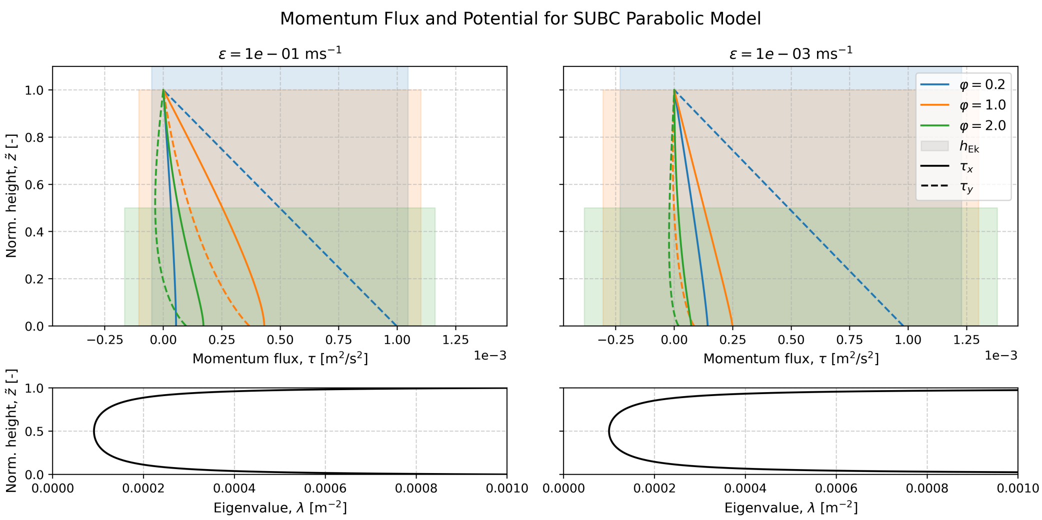

Figure 7: Numerically computed vertical structure for parabolic viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}} = 1/\tilde{h}_{\mathrm{Ek}}$ and the background viscosity ϵ. The corresponding Ekman heights and eigenvalue is shown with the shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow.

4.1.2 The surface angles

The surface angles from the numerical SUBC model can be viewed in Figure 8–10. There one can observe similarities between these models, the two-layer model, and the exponentially decaying model. They all have a lower boundary of 90° (except for the decreasing two-layer model, where the lower boundary was 0°).

The result from the linearly decreasing SUBC model is shown in Figure 8, where one can observe that the values go smoothly from 90° → 45° as φ increases. This plot resembles the exponential decaying model seen in Figure 4 and also the 1.5 model as seen in Figure 2.

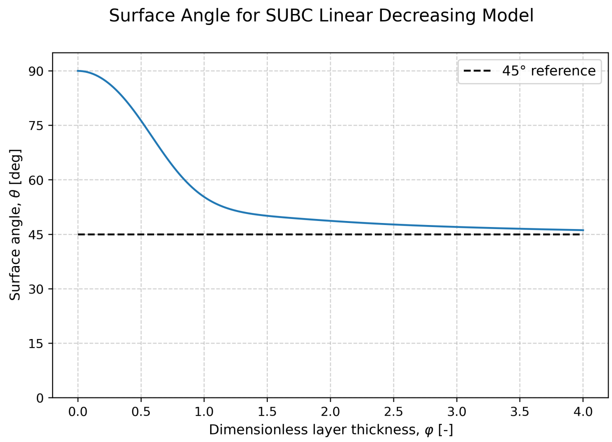

Figure 8: Numerically computed surface angles for decreasing viscosity profiles in the SUBC model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}}$. The solution assumed a purely zonal geostrophic flow.

However, for the linearly increasing viscosity a different result can be seen in Figure 9. The upper boundary is instead located at 0°, and the curve crosses 45°. One can observe different shapes for the different background viscosities, ϵ, although they all exhibit the same behaviour: an initial large angle decrease, a minimum, and then a slow increase. One can also observe that the lines show a sharper decrease and flatter behaviour for the smaller background viscosities.

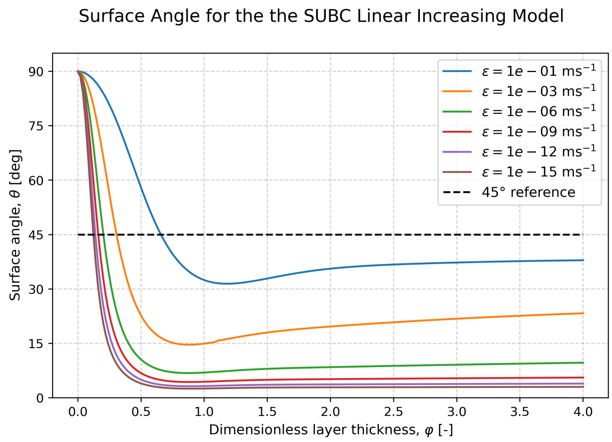

Figure 9: Numerically computed surface angles for linearly increasing. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}}$ and the background viscosity ϵ. The solution assumed a purely zonal geostrophic flow.

The SUBC model with a parabolic viscosity profile gave Figure 10. There one can observe very similar behaviour to the increasing viscosity profile in Figure 9. However, the two figures differ slightly in that the parabolic curves cross the 45° later than the linearly increasing curves. Furthermore, the minimum is less pronounced compared to the linear increasing plot.

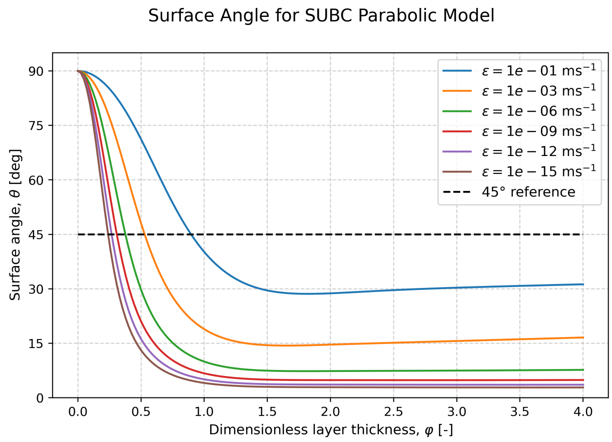

Figure 10: Numerically computed surface angles for a parabolic viscosity profile in the SUBC model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\text{Ek}}$ and the background viscosity ϵ. The solution assumed a purely zonal geostrophic flow.

4.1.3 The Ekman transport

Computing the transport as by the surface vertical momentum flux as Equation 12 yielded exactly the same shapes as seen in Figure 2–10, but centerd around 135° rather then 45°. The angle differences between the surface angles and the transport angle was thus always 90°, just like in the classical Ekman theory.

5 Discussion

6 Conclusion

A The Source Code

The source code for this article can be found on GitHub under:

`https://github.com/FredrikBergelv/Ekman-Spiral-Project`

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